

A practical RAG classroom handbook

Fictional teaching material - not real bookings or travel advice.

Ingestion and evidence

The fictional Atlas classroom collection contains travel guides, policy notes, diagrams and transcripts. Its reference code is ATLAS-LAB-73. Each upload is parsed into readable text before chunking. Image understanding should describe visible relationships and labels; it must not invent values obscured in a chart.

Token chunking divides text into manageable pieces. In this teaching example the chunk size is 512 tokens and the overlap is 50 tokens. Overlap repeats a small boundary region so a sentence near an edge has additional context. A token is not always a complete word.

Practice question

Which exact names, numbers and reference codes appear in this section?

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Three storage destinations

Vector storage holds embeddings that support similarity search by meaning. A BM25 corpus supports lexical matching, which is useful for exact references such as ATLAS-LAB-73. A graph stores extracted entities and relationships such as a coordinator leading a group. These representations serve different purposes.

Saving a graph does not prove that the answer pipeline queries it. Inspect retrieval separately from ingestion. In the classroom demonstration, verify vector and BM25 chunk identifiers after upload and report the graph extraction result separately. If graph extraction fails, do not describe it as a verified graph save.

Practice question

How does this section connect to the previous section?

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Evaluation and missing answers

Faithfulness asks whether the answer is supported by retrieved evidence. Relevance asks whether it addresses the question. Context precision concerns the usefulness of retrieved material. These concepts should be demonstrated with examples rather than interpreted as guarantees of correctness.

Ask one question with an exact reference, one paraphrased question, one question requiring two sections and one unanswerable question. The handbook does not specify the classroom Wi-Fi password. A grounded answer should acknowledge that missing information instead of inventing a password. Keep the test questions unchanged when comparing runs.

Practice question

What information is missing, and how should an assistant respond?